

SALT PILE STABILITY: A MATHEMATICAL MODEL

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Abstract—The maximum height for the salt pile in a circular dome with a 4 ft retaining wall was determined by two methods. The first method used rigid-body physics; in this model, the critical angle, the maximum angle of inclination allowed while maintaining static equilibrium, was determined using only the external coefficient of friction for salt. Because the static equilibrium also depended upon internal friction, a second model was developed. Development of the second model utilized particle physics, fluid mechanics and soil stress analysis. Mohr's circle, the internal coefficient of friction for salt and its angle of repose were used to determine the critical angle. These results were combined to form our solution model, Model II, which consisted of two submodels.

Model II(a) provides a general solution where the front-end loader is allowed to freely travel to any location on the salt pile. This model yields a maximum height of 17.4 ft for a symmetric cone with a critical angle of 14.6° .

Model II(b) provides a volume-maximizing solution if the loader's travel is restricted. This model yields a maximum height of 23.7 ft for a wedge shape with a ramp slope of 14.6° and a back edge slope of 35.9° , where the loader must not cross the peak.

Therefore, the authors recommend that Model II(a) be used in the general situation, since the loader is allowed to drive anywhere on the salt pile in this case. When the maximum volume provided is insufficient, Model II(b) can be utilized to increase the capacity of the dome. (Note: The loader must not cross the peak in this model.)

NOMENCLATURE

g = Gravity
 m = Mass of loader
 A = Surface area of loader on salt
 μ_e = External friction (salt on steel) = 0.627 [1, p. 9]
 μ_i = Internal friction (salt on salt) = 0.725 [1, p. 9]
 ϕ = Angle of repose of salt = critical angle of friction [2, p. 163], $\phi = \arctan \mu_i = 35.942^\circ$
 ρ = Bulk density of salt
 α = Angle of fracture with respect to the normal surface
 γ = Angle of fracture with respect to the ground
 β = Angle of inclination of the salt pile supporting the loader (angle of normal surface with respect to the ground)

δ = The supplement angle of $\beta + \phi$
 σ = Normal stress on the fault plane
 τ = Shear stress on the fault plane
 σ_n = Normal stress on the normal surface
 σ_s = Shear stress on the normal surface
 L = Arbitrary length along the fault plane
 τ_{max} = Critical salt shear stress
 $r = 0.5(\sigma_s - \sigma_n) \times$ critical salt shear stress
 h = Height of salt pile (ft)
 N = Normal force of loader on incline plane
 p = Stress exerted on salt pile by loader = mg/A
 d = Diameter of retaining wall = 103 ft
 w = Height of retaining wall = 4 ft

ASSUMPTIONS

1. The 4 ft retaining wall can withstand the pressure from any amount of salt that could be stored in the dome.
2. The ramp leading into the dome can be made at an angle flat enough to support the pressure of the salt in the dome, i.e. the salt ramp can provide as much support as the retaining wall.
3. The bottom 4 ft of the salt pile enclosed in the retaining wall can be ignored; therefore, the problem is to consider salt piled on a flat surface without walls.
4. The salt pile is homogeneous. *Definition of homogeneous*: salt particles have uniform size, shape, weight, density, internal and external coefficients of friction and temperature. Also these factors are independent of time.
5. The salt in the pile is uniformly packed.
6. The loader is supported by a continuous planar surface.
7. The acceleration of the loader has no effect on the salt pile.

8. The surface of the salt pile can be treated as an incline plane.
9. Salt has properties similar to soil:
 - (i) behaves similar to soil under stress;
 - (ii) fissures like soil;
 - (iii) supports objects in the same manner as soil;
 - (iv) compacts like soil.
10. The coefficient of internal friction tends to infinity in the first model (i.e. no relevant internal motion).
11. The coefficient of internal friction is some finite measurable number for the second model with the following two subcases (i.e. internal motion or fracture is possible):
 - (a) The front-end loader can safely travel on any portion of the salt pile:
 - (i) the shape of the salt pile will be conical with constant slope;
 - (ii) the rounding of the conical peak insignificantly damages the volume or height of the salt pile.
 - (b) The front-end loader will not drive beyond the crest of the peak at any time while constructing the salt pile (see Fig. 1):
 - (i) the shape of the salt pile will be a sliced cone;
 - (ii) the sliced cone can be treated as an infinite number of triangle slices;
 - (iii) the rounding of the salt pile crest will not significantly change the volume or height of the salt pile.

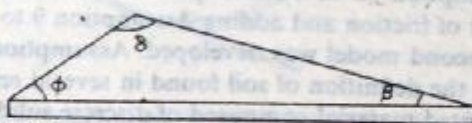
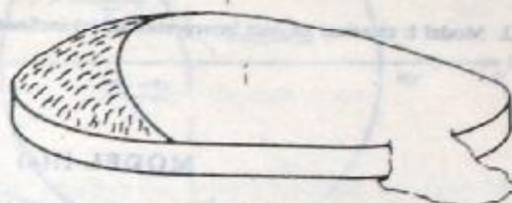
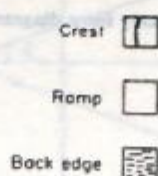


Fig. 1. Salt pile model design. Note: the loader must not leave the ramp; in particular, the loader must not cross the crest.

MOTIVATION OF THE MODEL

To better understand the problem, a preliminary model was constructed by considering the salt pile as an incline plane and applying classical physics to determine the height of the salt pile.

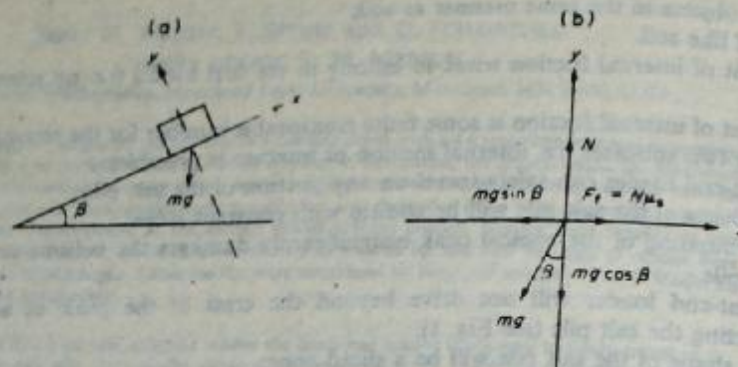
The situation appeared to be a static particle mechanics problem; therefore, fluid mechanics was researched. Information on piles and piling was also sought which produced references to soil mechanics. Since Mohr's circle seemed to be an important tool in solving fluid and soil mechanics problems, Mohr's circle was further investigated.

An in-depth analysis of Mohr's circle, as applied to soil, provided an understanding of the stress-strain relationships in soil. It was assumed that these relationships could be applied to the salt pile due to the similarities between salt and some types of soils (i.e. sand). Thus, Mohr's circle was used to analyze the salt pile and determine the maximum height of the salt pile.

MODEL I

Under Assumptions 1-8 and 10, the pile was considered as a perfect incline plane with coefficient of friction μ_s . Using particle dynamics [3, Chap. 6], the free-body force diagram was constructed and results were obtained (see Fig. 2). Assuming $\mu_s = 0.627$, the maximum height of the salt pile was determined to be 36.3 ft. Under these assumptions a 30 ft high salt pile is safe to work on as long as $\mu_s > 0.50$ (see Fig. 2).

Since the coefficient of internal friction for salt is a finite-measurable number, the stress caused by the loader may cause an internal fracture in the salt pile. Models II(a) and II(b) take internal friction into consideration.



$$\begin{aligned} \text{Sum of forces} = 0 \rightarrow & mg \sin \beta = N \mu_s \\ & mg \cos \beta = N \end{aligned}$$

$$\rightarrow \tan \beta = \mu_s \rightarrow \beta = \arctan \mu_s$$

$$\rightarrow \text{height (h)} = d \tan \beta / 2 \leq 30 \text{ ft}$$

$$\rightarrow \text{"safe" } \beta \text{ range: } \tan \beta \leq 0.5 < 0.583$$

Fig. 2. Model I: classical physics interpretation; (a) inclined plane, (b) free-body force diagram.

MODEL II(a)

Replacing Assumption 10 with Assumption 11(a), to take into consideration a finite measurable internal coefficient of friction and adding Assumption 9 to analyze the stress and fissure properties of salt piles, the second model was developed. Assumption 9 which indicates that salt is a type of soil, is justified by the definition of soil found in several references. For example, "soil ... is defined as any unconsolidated material composed of discrete solid particles with gases or liquids between" [4, pp. 2-3]. Assumption 11 follows from adoption of Assumption 9. Stress-strain analysis [5, Chap. 7; 2, Chap. 8] was applied to a small particle of the salt pile subjected to the weight of the loader (see Fig. 3). Mohr's circle was used to determine variable relationships (see Fig. 4).

Mohr's circle is a graphical representation of the static equilibrium conditions of a particle under stress. In the Problem Solution section, the static equilibrium equations are used to obtain the

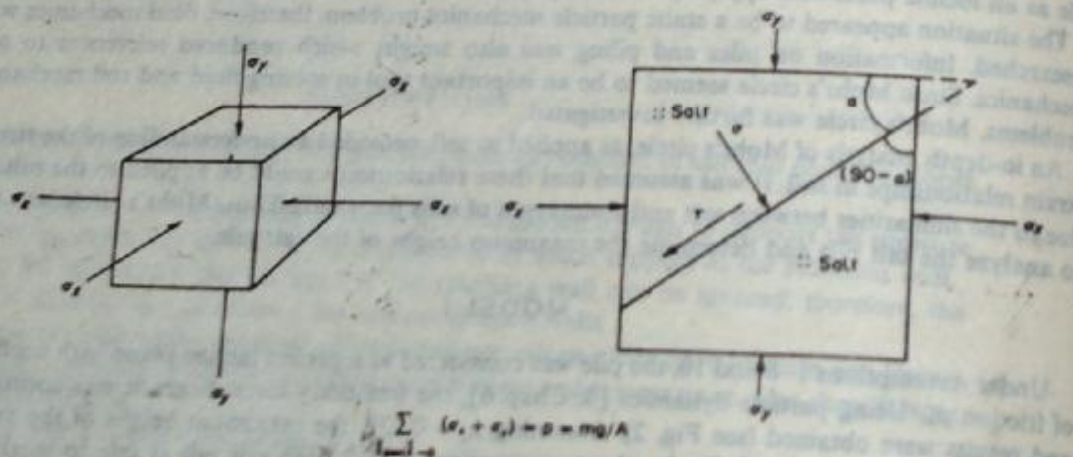


Fig. 3. Static stress and strain conditions for a small cube.

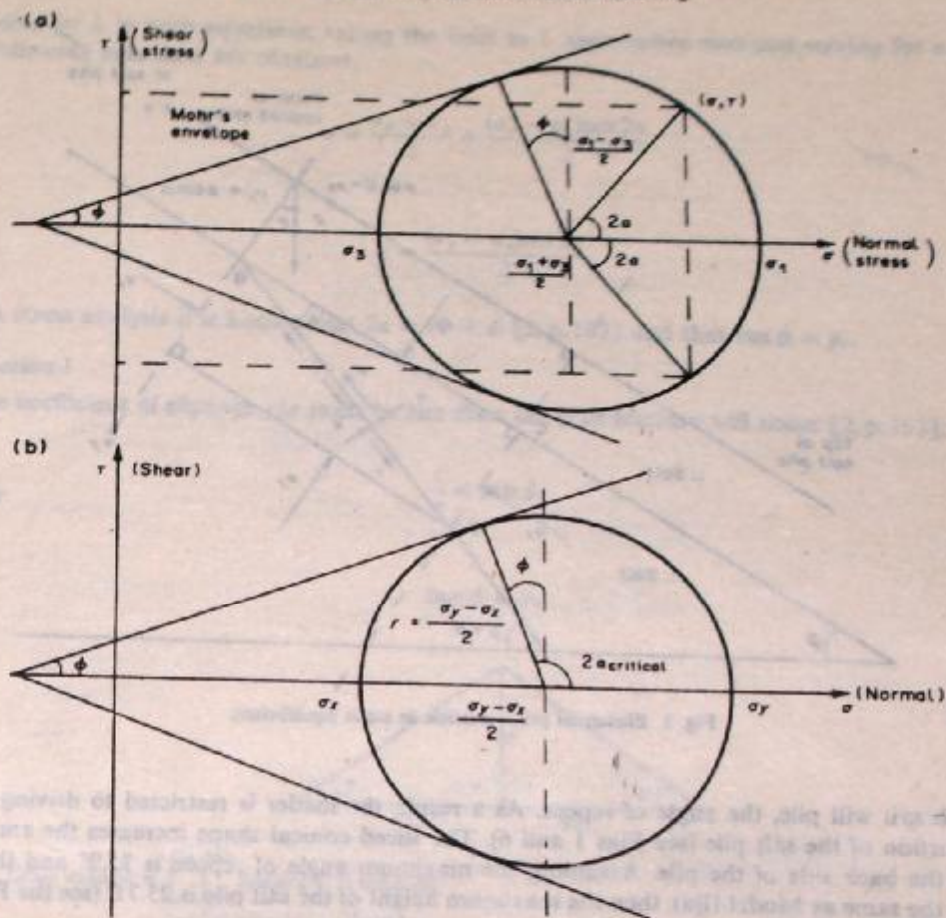


Fig. 4. Mohr's circle: (a) general Mohr's circle; (b) applied to the salt pile problem.

solution approximation. The maximum angle of inclination β of the salt pile that will support the loader is given by the following equation:

$$\beta = \text{arccot} \{ 2 / [\sin 2\alpha (\cot \phi - \cot 2\alpha) - 1] + 1 \},$$

where

$$\phi = \arctan \mu_r$$

and

$$2\alpha = 90 + \phi.$$

Assuming $\mu_r = 0.725$, then $\beta = 14.6^\circ$, and the maximum height of the salt pile is 17.4 ft (see the Problem Solution section).

MODEL II(b)

The second model may not utilize the storage facility as effectively as possible. In order to increase the efficiency of the storage facility the following model was developed. Assumption 11(a) was modified to Assumption 11(b) to allow the back of the pile to maintain the maximum angle

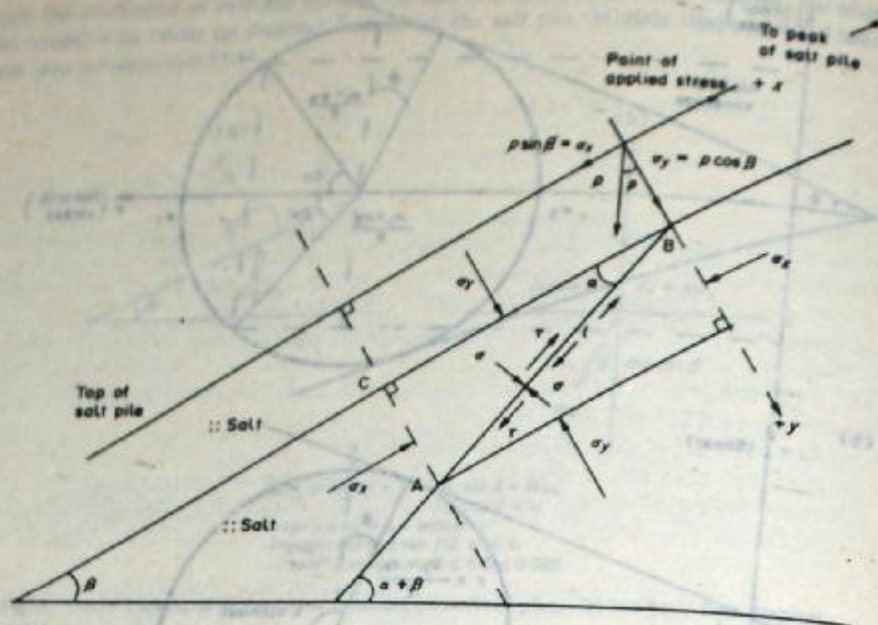


Fig. 5. Elemental prism particle in static equilibrium.

at which salt will pile, the angle of repose. As a result, the loader is restricted to driving on the front portion of the salt pile (see Figs 1 and 6). The sliced conical shape increases the amount of salt on the back side of the pile. Assuming the maximum angle of repose is 35.9° and the front angle is the same as Model II(a), then the maximum height of the salt pile is 23.7 ft (see the Problem Solution section below).

PROBLEM SOLUTION

In Fig. 5, an equilibrium prism was considered (triangle ABC). The solution method paralleled the sand-box example presented by Noble [2, p. 158]:

$$L = AB$$

$$L \sin \alpha = AC$$

$$L \cos \alpha = BC$$

$$p = mg/A$$

$$\sigma_x = p \sin \beta$$

$$\sigma_y = p \cos \beta.$$

Static equilibrium implies:

$$L \sigma_x \cos \alpha - L \tau \sin \alpha - L \sigma \cos \alpha = 0$$

and

$$L \sigma_y \sin \alpha - L \tau \sin \alpha - L \sigma \cos \alpha + L^2 \rho g \sin \alpha \cos \alpha = 0.$$

Dividing by L in both equations, taking the limit as L approaches zero and solving for σ and τ , the following equations are obtained:

$$\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{(\sigma_x - \sigma_y) \cos 2\alpha}{2}$$

and

$$\tau = \frac{(\sigma_x - \sigma_y) \sin 2\alpha}{2}$$

From stress analysis it is known that $2\alpha = 90^\circ + \phi$ [2, p. 163] and that $\tan \phi = \mu_f$.

Derivation I

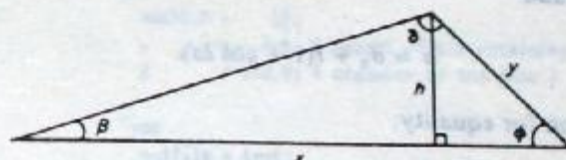
The coefficient of slippage τ/σ must be less than $\tan \phi$ or fracture will occur [2, p. 163]:

where
$$\frac{\tau}{\sigma} < \tan \phi,$$

$$\tan \phi = \mu_f$$



$$\begin{aligned} \text{Cone volume} &= \int_0^{2\pi} \int_0^{d/2} 2h(d/2 - r) r / d \, dr \, d\theta \\ &= 2\pi(2h)/d [d^2/4 - r^2/3]_0^{d/2} \\ &= \pi d^2 h / 12 \\ \text{Salt pile} &= \pi d^2 (h/3 + w) / 4 \\ \text{II(a) volume} & \end{aligned}$$



Law of sines:

$$\begin{aligned} \sin \beta / y &= \sin \delta / x \\ \sin \phi &= h / y \\ x \sin \beta / \sin \delta &= y \\ \rightarrow h &= x \sin \beta \sin \phi / \sin \delta \end{aligned}$$

$$\text{Area of triangle} = h/2 = \frac{x^2 \sin \beta \sin \phi}{2 \sin \delta}$$

$$\begin{aligned} \text{Wedge volume} &= \int \text{Area} \, d(\text{Area}) \\ &= 2 \int_0^x \frac{x^2 \sin \beta \sin \phi}{2 \sin \delta} \, dx \\ &= \frac{d^3 \sin \beta \sin \phi}{3 \sin \delta} \end{aligned}$$

$$\begin{aligned} \text{Salt pile} &= \frac{d^3 \sin \beta \sin \phi}{3 \sin \delta} + \frac{\pi d^2 w}{4} \\ \text{II(b) volume} & \end{aligned}$$

Fig. 6. Salt pile volumes.

Solving the critical equation for equality:

$$\begin{aligned}\frac{1}{\tan \phi} &= \frac{\sigma}{\tau} \\ &= \left[\frac{\sigma_x + \sigma_z}{2} + \frac{(\sigma_y - \sigma_x) \cos 2\alpha}{2} \right] \frac{(\sigma_y - \sigma_x) \sin 2\alpha}{2} \\ &\rightarrow \cot \phi - \cot 2\alpha = \frac{\sigma_x + \sigma_z}{(\sigma_y - \sigma_x) \sin 2\alpha} \\ &\rightarrow \sin 2\alpha (\cot \phi - \cot 2\alpha) = \frac{\sigma_x + \sigma_z}{\sigma_y - \sigma_x} \\ &= \frac{p \sin \beta + p \cos \beta}{p \cos \beta - p \sin \beta} = \frac{1 + 2 \sin \beta}{\cos \beta - \sin \beta} \\ &\rightarrow \frac{1}{\sin 2\alpha (\cot \phi - \cot 2\alpha) - 1} = \frac{\cot \beta - 1}{2} \\ &\rightarrow \cot \beta = 1 + \frac{2}{\sin 2\alpha (\cot \phi - \cot 2\alpha) - 1}\end{aligned}$$

Thus,

$$\beta = \operatorname{arccot} \left(1 + \frac{2}{\sin 2\alpha (\cot \phi - \cot 2\alpha) - 1} \right).$$

The β calculated here is beta1 in the program.

Derivation II

Mohr's circle provided a second, equivalent equation. Define r = radius of Mohr's circle. From Mohr's circle, $\tau = r \sin 2\alpha$ and

$$\sigma = \sigma_x + r(1 + \cos 2\alpha).$$

Solving the critical equation for equality:

$$\begin{aligned}\frac{\tau}{\sigma} &= \frac{r \sin 2\alpha}{\sigma_x + r(1 + \cos 2\alpha)} \\ &= \tan \phi\end{aligned}$$

$$\rightarrow \frac{\sigma_x}{r} = \frac{\sin 2\alpha}{\tan \phi} - (1 + \cos 2\alpha), \quad (1)$$

$$\frac{\sigma_z}{\sigma_x} = \frac{p \cos \beta}{p \sin \beta} = \cot \beta \quad (2)$$

and

$$\frac{\sigma_x}{r} = \frac{\sigma_x}{(\sigma_y - \sigma_x)/2} = \frac{2}{\sigma_y/(\sigma_x - 1)}, \quad (3)$$

$$\text{equations (1)-(3)} \rightarrow \frac{2}{\cot \beta - 1} = \frac{\sin 2\alpha}{\tan \phi - (1 + \cos 2\alpha)}$$

$$\rightarrow \cot \beta = \frac{2 \tan \phi}{\sin 2\alpha - (1 + \cos 2\alpha) \tan \phi} + 1.$$

Thus

$$\beta = \arctan \left(\frac{1}{\{2 \tan \phi / [\sin 2\alpha - (1 + \cos 2\alpha) \tan \phi]\} + 1} \right)$$

The β calculated here is beta2 in the program.

Salt pile height

In each previous development, the maximum possible β was derived. To find the maximum salt height h given the maximum angle of inclination β , we use the following equations:

Method II(a),

$$h = \frac{w + d \tan \beta}{2};$$

Method II(b),

$$h = \frac{w + d \sin \beta \sin \phi}{\sin \delta}.$$

The maximum height occurs at the center of the dome for Method II(a) and at the crest peak for Method II(b).

PROGRAM MODEL

```
( Program to determine the maximum height of the salt pile
and the volume of the salt pile for all three models.

The program input is the Angle of Repose (in degrees). )
```

```
Program Model (input,output);
```

```
Const
```

```
  namelen = 12;
```

```
  w      = 4.0; { height of the retaining wall }
```

```
  d      = 103.0; { diameter of the dome }
```

```
Type
```

```
  outfile = text;
```

```
Var
```

```
  wu, phi, alpha : real;
```

```
  beta1, beta2   : real;
```

```
  vol, vola, volb : real;
```

```
  h, ha, hb       : real;
```

```
  filename        : string(namelen);
```

```
  dat             : outfile;
```

```
Function pi : real;
```

```
begin
```

```
  pi := 4.0 * arctan(1);
```

```
end;
```

```
Function cotan(theta : real) : real;
```

```
begin
```

```
  cotan := cos(theta) / sin(theta);
```

```
end;
```

```

Function tan( theta : real ) : real;
begin
  tan := sin( theta )/cos( theta );
end;

Function arccotan( lambda : real ) : real;
begin
  arccotan := arctan(1.0/lambda);
end;

Function rad( degree : real ) : real;
begin
  rad := degree*pi/180.0;
end;

Function deg( radian : real ) : real;
begin
  deg := radian*180.0/pi;
end;

Function min( a, b, c : real ) : real;
begin
  function mint( a, b : real ) : real;
  begin
    if (a < b) then
      mint := a;
    else
      mint := b;
    end;
  end;

  if (a < b) then
    min := mint(a, c);
  else
    min := mint(b, c);
  end;
end;

Begin
  writeln;
  writeln('Enter name for output file. ');
  readln(filename);
  assign(dat,filename);
  writeln;
  rewrite(dat);
  writeln('Enter the repose angle. ');
  read( phi );
  while (phi > 0) do
    begin
      phi := rad(phi);
      alpha := (rad(90) + phi)/2.0;
      writeln;
      writeln;
      writeln(dat);
      writeln(dat);
      writeln(dat, ' Angle of Repose = ', deg(phi):6:3);
      writeln(dat);
      beta1 := arccotan( 2.0/( sin(2.0*alpha)*( cotan(phi)-cotan(2.0*alpha) ) - 1.0) + 1.0 );
      beta2 := arccotan( 2.0*tan(phi)/(sin(2*alpha)-(1+cos(2*alpha))*tan(phi)) + 1);
      writeln(dat, 'Critical Angle (Beta1) = ', deg(beta1):6:3);
      writeln(dat, 'Critical Angle (Beta2) = ', deg(beta2):6:3);
      writeln(dat);
      h := d * tan(phi) / 2.0;
      vol := pi * d * d * (h / 12.0 + w / 4.0);
    end;
  end;

```

```

ha := d * tan(beta1) / 2.0;
vol := pi * d * d * ha / 12.0 + w / 4.0;
hb := d * sin(beta1) * sin(phi) / sin(phi - beta1 - phi1);
volb := d * d * (hb / 2.0 + w * pi / 4.0);
n := h + 4.0;
ha := ha + 4.0;
hb := hb + 4.0;
writeln(dat, 'Model I      Maximum Height of Salt Pile (h) = ', hc6:3);
writeln(dat, 'Model I      Volume Yield = ', vol:10:3);
writeln(dat);
writeln(dat, 'Model II(a) Maximum Height of Salt Pile (h) = ', ha6:3);
writeln(dat, 'Model II(a) Volume Yield = ', vola:10:3);
writeln(dat);
writeln(dat, 'Model II(b) Maximum Height of Salt Pile (h) = ', hb6:3);
writeln(dat, 'Model II(b) Volume Yield = ', volb:10:3);
writeln(dat);
write(dat, 'Model Solution: Maximum Height of Salt Pile (h) = ');
writeln(dat, min(h, ha, hb):6:3);
writeln(dat);
writeln('Enter the repose angle. ');
readln(phi);
end;
close(dat);
writeln;
End.

```

PROGRAM OUTPUT

Angle of repose = 10.000

Critical angle $\beta_1 = 35.149$
 Critical angle $\beta_2 = 35.149$

Model I	Maximum height of salt pile (h) = 13.081
Model I	Volume yield = 58550.550
Model II(a)	Maximum height of salt pile (h) = 40.261
Model II(a)	Volume yield = 134040.262
Model II(b)	Maximum height of salt pile (h) = 18.524
Model II(b)	Volume yield = 84691.961
Model solution	Maximum height of salt pile (h) = 13.081

Angle of repose = 15.000

Critical angle $\beta_1 = 30.489$
 Critical angle $\beta_2 = 30.489$

Model I	Maximum height of salt pile (h) = 17.799
Model I	Volume yield = 71655.974
Model II(a)	Maximum height of salt pile (h) = 34.323
Model II(a)	Volume yield = 117548.384
Model II(b)	Maximum height of salt pile (h) = 22.967
Model II(b)	Volume yield = 100403.252
Model solution	Maximum height of salt pile (h) = 17.799

Angle of repose = 20.000

Critical angle $\beta_1 = 26.118$
 Critical angle $\beta_2 = 26.118$

Model I	Maximum height of salt pile (h) = 22.744
Model I	Volume yield = 85390.596
Model II(a)	Maximum height of salt pile (h) = 29.250
Model II(a)	Volume yield = 103459.161
Model II(b)	Maximum height of salt pile (h) = 25.516
Model II(b)	Volume yield = 109417.715
Model solution	Maximum height of salt pile (h) = 22.744

Angle of repose = 25.000

Critical angle $\beta_1 = 22.090$

Critical angle $\beta_2 = 22.090$

Model I Maximum height of salt pile (h) = 26.018

Model I Volume yield = 100026.698

Model II(a) Maximum height of salt pile (h) = 24.902

Model II(a) Volume yield = 81382.197

Model II(b) Maximum height of salt pile (h) = 26.350

Model II(b) Volume yield = 112367.602

Model solution Maximum height of salt pile (h) = 24.902

Angle of repose = 30.000

Critical angle $\beta_1 = 18.435$

Critical angle $\beta_2 = 18.435$

Model I Maximum height of salt pile (h) = 33.734

Model I Volume yield = 118911.971

Model II(a) Maximum height of salt pile (h) = 21.167

Model II(a) Volume yield = 81008.366

Model II(b) Maximum height of salt pile (h) = 25.766

Model II(b) Volume yield = 110302.619

Model solution Maximum height of salt pile (h) = 21.167

Angle of repose = 35.000

Critical angle $\beta_1 = 15.162$

Critical angle $\beta_2 = 15.162$

Model I Maximum height of salt pile (h) = 40.061

Model I Volume yield = 133485.183

Model II(a) Maximum height of salt pile (h) = 17.956

Model II(a) Volume yield = 72090.932

Model II(b) Maximum height of salt pile (h) = 24.124

Model II(b) Volume yield = 104493.583

Model solution Maximum height of salt pile (h) = 17.956

Angle of repose = 35.942

Critical angle $\beta_1 = 14.589$

Critical angle $\beta_2 = 14.589$

Model I Maximum height of salt pile (h) = 41.337

Model I Volume yield = 137031.012

Model II(a) Maximum height of salt pile (h) = 17.404

Model II(a) Volume yield = 70557.053

Model II(b) Maximum height of salt pile (h) = 23.726

Model II(b) Volume yield = 103086.905

Model solution Maximum height of salt pile (h) = 17.404

Angle of repose = 40.000

Critical angle $\beta_1 = 12.268$

Critical angle $\beta_2 = 12.268$

Model I Maximum height of salt pile (h) = 47.214

Model I Volume yield = 153351.979

Model II(a) Maximum height of salt pile (h) = 15.198

Model II(a) Volume yield = 64431.664

Model II(b) Maximum height of salt pile (h) = 21.787

Model II(b) Volume yield = 96230.814

Model solution Maximum height of salt pile (h) = 15.198

Angle of repose = 45.000

Critical angle $\beta_1 = 9.736$

Critical angle $\beta_2 = 9.736$

Model I Maximum height of salt pile (h) = 55.500

Model I Volume yield = 176366.786

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Model II(a)	Maximum height of salt pile (h) = 12.836
Model II(a)	Volume yield = 57870.534
Model II(b)	Maximum height of salt pile (h) = 19.084
Model II(b)	Volume yield = 86671.211
Model solution	Maximum height of salt pile (h) = 12.836

Angle of repose = 50.000

Critical angle $\beta_1 = 7.546$
Critical angle $\beta_2 = 7.546$

Model I	Maximum height of salt pile (h) = 86.375
Model I	Volume yield = 202754.766

Model II(a)	Maximum height of salt pile (h) = 10.822
Model II(a)	Volume yield = 52277.871

Model II(b)	Maximum height of salt pile (h) = 16.280
Model II(b)	Volume yield = 76754.762

Model solution	Maximum height of salt pile (h) = 10.822
----------------	--

Angle of repose = 55.000

Critical angle $\beta_1 = 5.677$
Critical angle $\beta_2 = 5.677$

Model I	Maximum height of salt pile (h) = 77.550
Model I	Volume yield = 237608.062

Model II(a)	Maximum height of salt pile (h) = 9.120
Model II(a)	Volume yield = 47549.003

Model II(b)	Maximum height of salt pile (h) = 13.573
Model II(b)	Volume yield = 67183.125

Model solution	Maximum height of salt pile (h) = 9.120
----------------	---

Angle of repose = 60.000

Critical angle $\beta_1 = 4.107$
Critical angle $\beta_2 = 4.107$

Model I	Maximum height of salt pile (h) = 93.201
Model I	Volume yield = 281077.599

Model II(a)	Maximum height of salt pile (h) = 7.698
Model II(a)	Volume yield = 43598.796

Model II(b)	Maximum height of salt pile (h) = 11.101
Model II(b)	Volume yield = 58439.700

Model solution	Maximum height of salt pile (h) = 7.698
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WEAKNESSES OF THE MODEL

Although the development of the model considered the weight (mass) of the loader, the proposed model is independent of that parameter. Miniature scale model experimentation tended to support this result. Also, since loaders have been used to pile salt for 15 years, the salt must be able to support the weight of the loader.

Lack of sample data and estimated constants (internal coefficient of friction for road salt and angle of repose for road salt) despite an exhaustive search, rendered error analysis inappropriate. The error is then assumed to be due primarily to measurement of the parameters of the model. Normal error analysis could then be computed for the parameters, thus giving appropriate bounds for the maximum height (see the Appendix for a possible method).

STRENGTHS OF THE MODEL

The model appears reasonable since the macroscopic properties of salt and soil (specifically sand) are similar. The results obtained were within the range of the height suggested by the panel to the actual height used in the past, which indicated that the model is realistic. The model is general enough to be applied to soil, sand and other flowing solids with a measurable coefficient of internal friction. The critical variable needed to apply the model is the coefficient of internal friction, which

can be experimentally obtained. Thus the model is easy to implement and maintain because the only independent variables needed to determine the maximum angle of inclination of the salt pile are the internal coefficient of friction and angle of repose. In Model II(a), the loader needs to restrict the angle on all sides of the pile, keeping it below the maximum angle of inclination. Similarly, in Model II(b), only the front side of the sliced conical shape must be maintained below the maximum angle of inclination, while the back side of the pile can be at the angle of repose.

TESTING THE MODEL

To better understand the angle of repose and how salt behaves when piled, a miniature model of the situation was constructed. The miniature model consisted of table salt and a cardboard ring of 6" dia. Rough experimentation indicated that the angle of repose (35.9°) appeared accurate and the stress caused by objects on the angle surface did cause faults to occur. Therefore, a scale model could be used to experimentally check the validity of the model: (1) materials with varying degrees of homogeneity, as defined in the model, could be tested—if this testing indicates that the model is poor, it could be conditioned to allow for variables which appear to affect the maximum height of the salt pile; (2) the effects of varying stress, e.g. changes in the weight of the loader, could be studied; (3) the relationship between the internal coefficient of friction and maximum height of the salt pile could be tested and the predicted relationship between the angle of repose and maximum height could be tested (see the Appendix).

CONCLUSIONS

Given a coefficient of internal friction, three situations are possible:

- (1) The coefficient of internal friction is very small, indicating the salt will not support the loader
- (2) The salt will support the loader and the coefficient of external friction is small, indicating the loader will slide rather than cause internal stress fractures.
- (3) The salt will support the loader and the coefficient of external friction is large, indicating the loader will cause stress fractures before sliding.

It should be noted that the coefficient of internal friction should be > 0.5 , since materials with smaller coefficients would probably not pile up as they would flow away too easily.

The results of the model indicated that Model II(a) should be used in the general situation since the loader is allowed to drive anywhere on the salt pile, but when the maximum volume provided by Model II(a) is insufficient, Model II(b) can be utilized to increase the capacity of the dome. (Note: the loader must not cross the peak in this situation.)

REFERENCES

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6. F. R. Giordano and M. D. Weir, *A First Course in Mathematical Modeling*. Brooks/Cole, Monterey, Calif. (1985).

See overleaf for the Appendix

APPENDIX

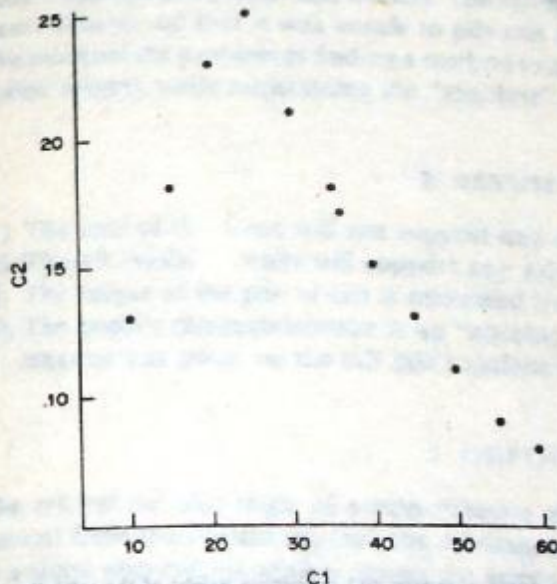
Error Analysis Example

The following analysis was done using Minitab on the Vax 11/780 computer by the graphical curve-fitting technique [6, pp. 09-96].

MTB > note repose angle in C1 max h in C2
MTB > print C1 C2

ROW	C1	C2
1	10.000	13.081
2	15.000	17.799
3	20.000	22.740
4	25.000	24.902
5	30.000	21.167
6	35.000	17.956
7	35.942	17.404
8	40.000	15.198
9	45.000	12.863
10	50.000	10.822
11	55.000	9.120
12	60.000	7.698

MTB > plot C2 vs C1



MTB > regress max h in C2 using 1 pred repose angle in C1

The regression equation is
 $C2 = 24.0 - 0.231 C1$

Predictor	Coef	Stdev	t-ratio
Constant	24.003	3.051	7.87
C1	-0.23111	0.07986	-2.89
s	4.188	R-sq = 45.6%	R-sq(adj) = 40.1%

Analysis of Variance

Source	DF	SS	MS
Regression	1	146.93	146.93
Error	10	175.42	17.54
Total	11	322.35	

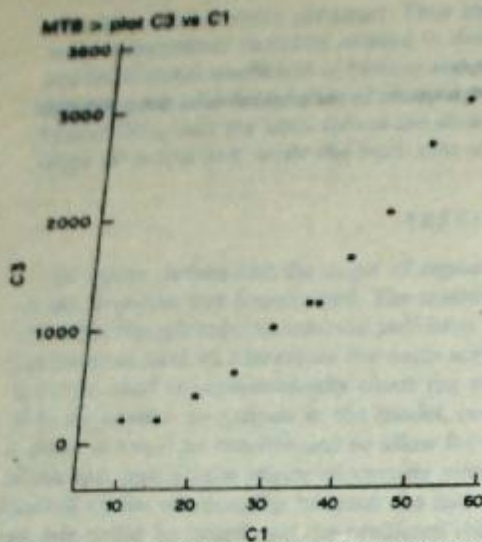
Unusual Observations

Obs.	C1	C2	Fit	St. Fit	Resid.	St. Resid.
1	10.0	13.08	21.69	2.34	-8.61	-2.48R

R denotes an Obs. with a large St. Resid.

MTB > mult C1 by C1 put in C3

MTB > note max height vs repose angle square



MTB > note max height vs angle of repose and angle squared
MTB > regress C2 using 2 pred C1 C3

The regression equation is

$$C2 = 11.0 + 0.685 C1 - 0.0131 C3$$

Predictor	Coef	Stdev	t-ratio
Constant	10.993	4.236	2.60
C1	0.6849	0.2653	2.58
C3	-0.013101	0.003714	-3.53

s = 2.860 R-sq = 77.2% R-sq(adj) = 72.1%

Analysis of Variance

Source	DF	SS	MS
Regression	2	248.72	124.36
Error	9	73.62	8.18
Total	11	322.35	

Source	DF	Seq SS
C1	1	146.93
C3	1	101.80

MTB > stop

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Using this analysis, there is a quadratic relationship between the angle of repose and maximum height of the salt pile. This would also indicate that the error involved in the predicted maximum height has a quadratic relationship to the error in the determination of the angle of repose.